

```
data = read.csv("data/expenditure.csv", sep=",")

dim(data)
```

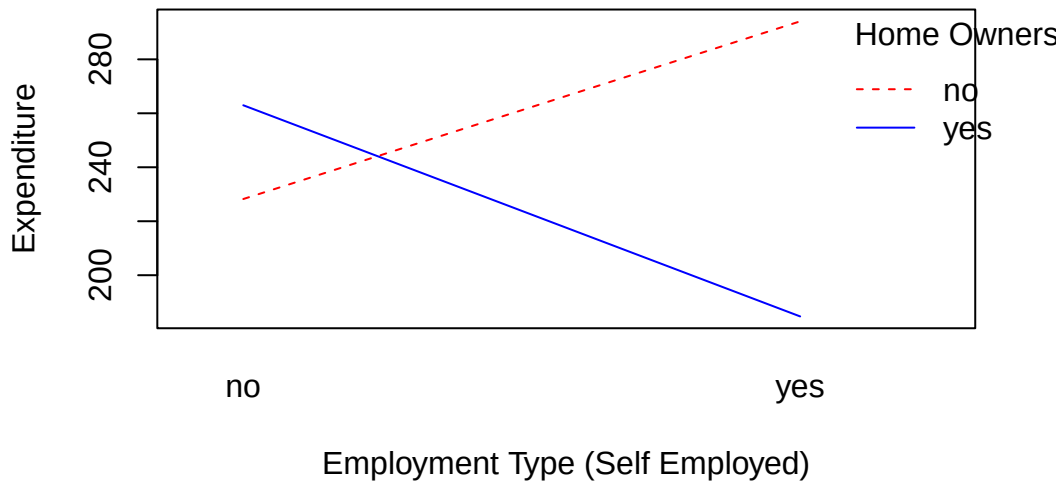
```
[1] 1002    6
```

```
head(data)
```

	age	income	expenditure	homeowner	selfemp	dependents
1	37.66667	4.5200	124.983300	yes	no	3
2	33.25000	2.4200	9.854167	no	no	3
3	33.66667	4.5000	15.000000	yes	no	4
4	30.50000	2.5400	137.869200	no	no	0
5	32.16667	9.7867	546.503300	yes	no	2
6	23.25000	2.5000	91.996670	no	no	0

i

```
interaction.plot(x.factor = data$selfemp, trace.factor = data$homeowner, response=data$expenditure)
```



Since the lines are not parallel, there is most likely an interaction affect.

ii

```
model_full <- lm(expenditure ~ age + income + homeowner + selfemp + dependents +  
                 homeowner*selfemp, data = data)  
summary(model_full)
```

Call:

```
lm(formula = expenditure ~ age + income + homeowner + selfemp +  
    dependents + homeowner * selfemp, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-679.12	-146.51	-64.23	71.83	2372.32

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	130.3112	31.2636	4.168	3.34e-05 ***
age	-2.2948	0.9679	-2.371	0.0179 *
income	58.3747	5.7728	10.112	< 2e-16 ***
homeowneryes	-9.6305	20.2774	-0.475	0.6349
selfempyes	18.7806	58.9144	0.319	0.7500
dependents	-4.7977	7.7030	-0.623	0.5335
homeowneryes:selfempyes	-125.8407	75.1981	-1.673	0.0946 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 274.4 on 995 degrees of freedom

Multiple R-squared: 0.1015, Adjusted R-squared: 0.09612

F-statistic: 18.74 on 6 and 995 DF, p-value: < 2.2e-16

iii

According to the summary in part ii, the interaction effect coefficient had a p-value of 0.0946 which is less than $\alpha = 0.10$ so the interaction effect is significant.

iv

I will use a reduced model based on the coefficients that had a p-value less than 0.1. Thus:

```
model_red <- lm(expenditure ~ age + income + homeowner*selfemp, data = data)
summary(model_red)
```

Call:

```
lm(formula = expenditure ~ age + income + homeowner * selfemp,
    data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-673.00	-146.98	-62.55	70.67	2382.59

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	131.4656	31.1990	4.214	2.74e-05 ***
age	-2.3362	0.9653	-2.420	0.0157 *
income	57.4764	5.5880	10.286	< 2e-16 ***
homeowneryes	-12.4917	19.7441	-0.633	0.5271
selfempyes	17.1037	58.8348	0.291	0.7713
homeowneryes:selfempyes	-122.9871	75.0354	-1.639	0.1015

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 274.3 on 996 degrees of freedom

Multiple R-squared: 0.1012, Adjusted R-squared: 0.09667

F-statistic: 22.43 on 5 and 996 DF, p-value: < 2.2e-16

V

```
anova(model_full, model_red, test="F")
```

Analysis of Variance Table

Model 1: expenditure ~ age + income + homeowner + selfemp + dependents +
homeowner * selfemp

Model 2: expenditure ~ age + income + homeowner * selfemp

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	995	74911690				
2	996	74940896	-1	-29205	0.3879	0.5335

Here the F-test results in a $p\text{-val} = 0.5335 > \alpha = 0.1$ so there is no significant relation between expenditure and the extra parameters.

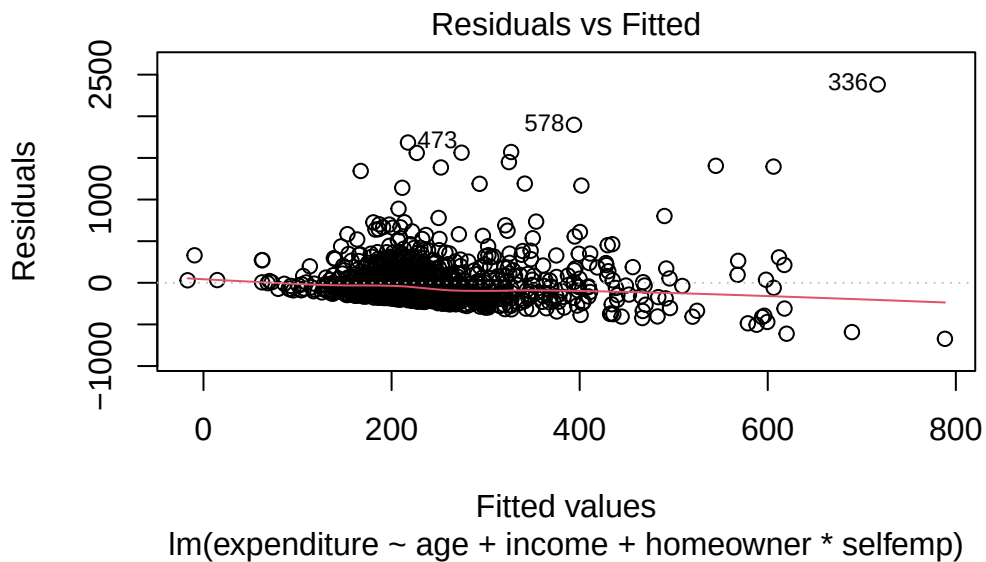
```
summary(model_red)$r.squared
```

```
[1] 0.1011848
```

```
summary(model_red)$adj.r.squared
```

```
[1] 0.09667272
```

```
plot(model_red, which=1)
```



However, both the R^2 and R^2_{adj} are rather small so the model may not be the best fit. Additionally, looking at the residual vs fitted plot has a changing spread which could indicate non constant variance and poor fit.

vi

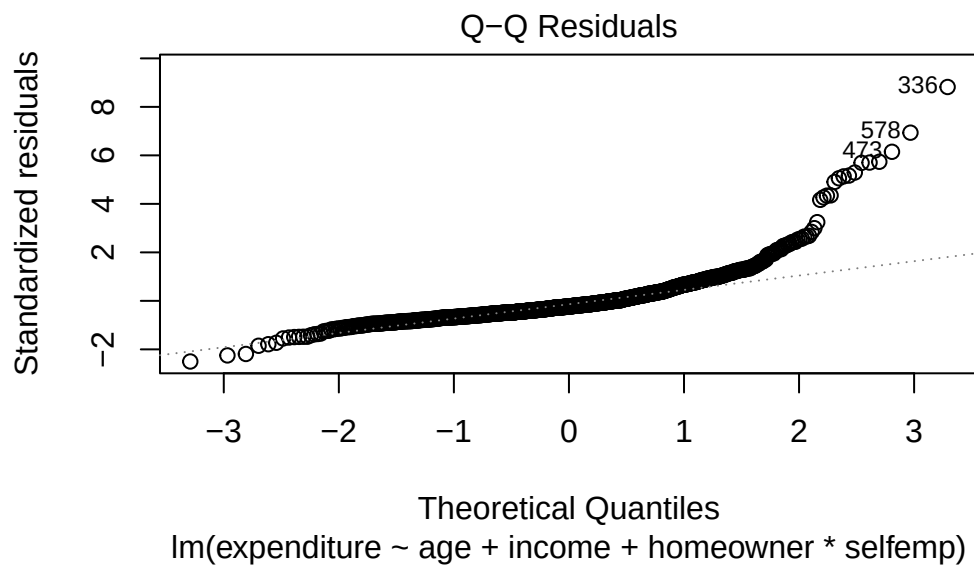
```
ncvTest(model_red)
```

Non-constant Variance Score Test
Variance formula: $\sim \text{fitted.values}$
Chisquare = 495.9129, Df = 1, p = $< 2.22\text{e-}16$

Here the p-value is less than any reasonable α so we reject the null hypothesis that there is constant variance of the residuals. We also noted that the constant variance assumption may not be satisfied in part v.

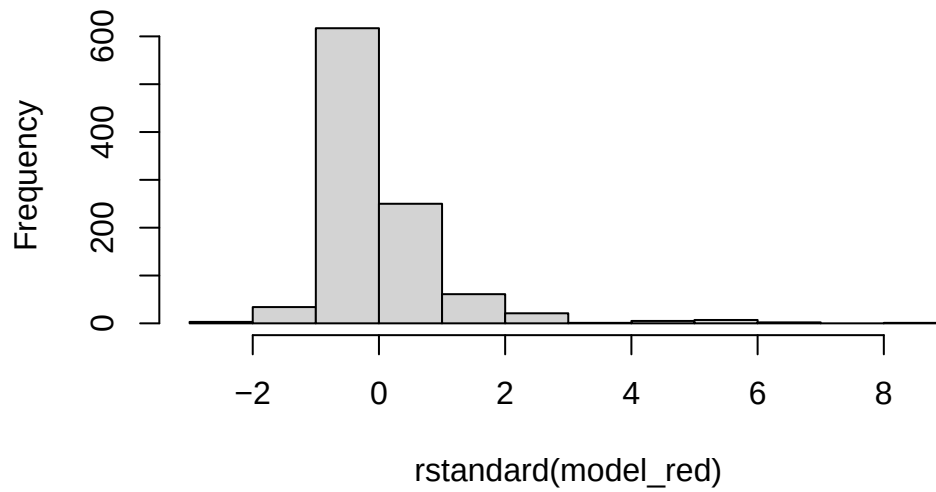
vii

```
plot(model_red, which=2)
```



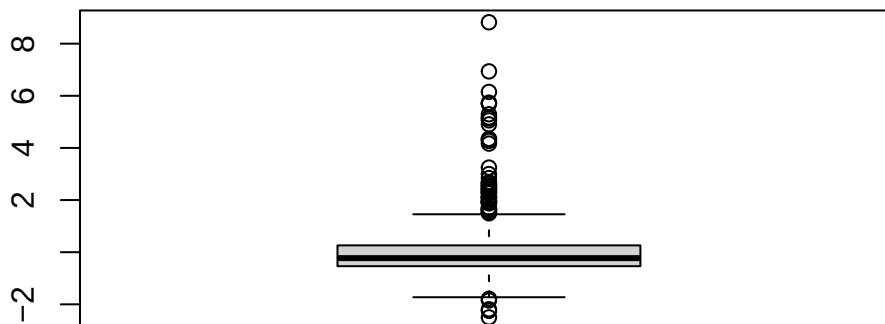
```
hist(rstandard(model_red), main="Histogram of standardized residuals")
```

Histogram of standardized residuals



```
boxplot(rstandard(model_red), main="Boxplot of standardized residuals")
```

Boxplot of standardized residuals



```
## Shapiro-Wilk's test  
shapiro.test(model_red$residuals)
```

Shapiro-Wilk normality test

data: model_red\$residuals
W = 0.74944, p-value < 2.2e-16

Here the p-value is less than any reasonable α so we reject the null hypothesis that the residuals are normally distributed.

It is also clear from the QQ plot and the box plot that this distribution is likely not normal.

viii

```
durbinWatsonTest(model_red)
```

```
lag Autocorrelation D-W Statistic p-value
1   -0.005070022      2.009564    0.93
Alternative hypothesis: rho != 0
```

Here the p-value (0.946) is greater than any reasonable α so we do not reject the null hypothesis and the independence assumption is satisfied.

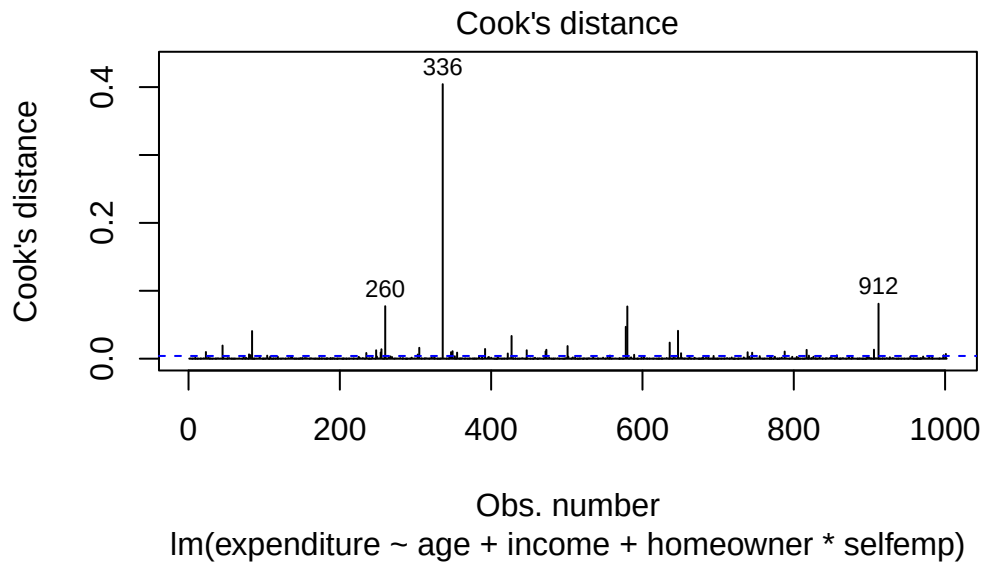
ix

```
## Cook's distance
D <- cooks.distance(lm(model_red))
cutoff1 <- qf(0.5, 4, df.residual(model_red))
print(cutoff1)
```

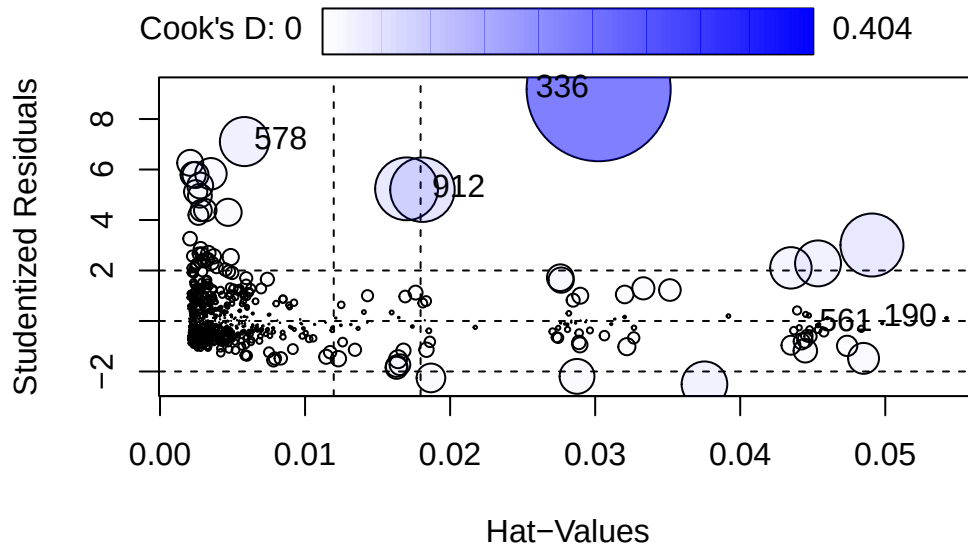
```
[1] 0.8397453
```

```
cutoff2 <- 4/df.residual(model_red)
cutoff3 <- 0.5

plot(model_red, which=4)
abline(h=cutoff1, lty=2, col="black")
abline(h=cutoff2, lty=2, col="blue")
abline(h=cutoff3, lty=2, col="red3")
```



```
## influence plot
influencePlot(model_red)
```



	StudRes	Hat	CookD
190	0.11321809	0.05422846	1.226174e-04
336	9.18185637	0.03023961	4.043301e-01
561	-0.09526826	0.04979745	7.935393e-05
578	7.10698893	0.00582018	4.694863e-02
912	5.20042607	0.01809249	8.093641e-02

$F(0.5, 4, 998) = 0.8397453$ and none of the points have a cooks distance above this point. Therefore none of the points are influential (although we have a few outliers.)

x

```
vif(model_red)
```

there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif

	age	income	homeowner	selfemp
	1.263228	1.219949	1.296166	2.595034
homeowner:selfemp				
	2.666452			

Here all the VIF scores are less than 10 so we do not believe there is any multicollinearity.